

THORACIC TRAUMA

Introduction & epidemiology

- **Second leading cause of trauma deaths.** 65–70% of chest injuries from RTAs. **~25% of trauma deaths due to chest injuries.** 50% of polytrauma deaths have significant chest injury.
- Immediate death = rupture of myocardium / thoracic aorta. **Early deaths (30 min–3h) =** tension pneumothorax, cardiac tamponade, airway obstruction, uncontrolled haemorrhage — often **preventable**.

Rib fractures

- Heal 3–6 weeks. Ribs 4–9 commonest.
- **Lower ribs (X–XII) → intra-abdominal injury (spleen, liver, kidney).**
- **1st + 2nd rib fracture + apical cap → great vessel injury** (high energy).
- Management: pain control, pulmonary function. **Elderly + multiple rib # → admit** (pneumonia risk).

Flail chest

- **≥3 consecutive ribs fractured in ≥2 places** → free segment → **paradoxical respiration** (moves IN on inspiration, opposite to normal).
- Associated **pulmonary contusion = the real problem**.
- Management: pain control, pulmonary toilet, O₂, treat contusion. Severe → intubation + IPPV (internal pneumatic fixation).

Sternal fracture

- Anterior blunt / **seatbelt deceleration**. Low mortality.
- **Screen for myocardial contusion (ECG + serial cardiac enzymes), monitor 24–48h.**
- NO association with blunt aortic injury.

Subcutaneous emphysema

- Air in chest wall = marker of serious injury. Benign/self-limiting → **high-flow O₂**. Massive → "gills" incisions.

Pulmonary contusion

- Alveolar oedema/haemorrhage; hypoxaemia ~50%; **X-ray findings within minutes, ARDS-like course.**
- Management: **restrict IV fluids**, pulmonary toilet, respiratory support. **NO prophylactic antibiotics.** Mortality 5–16%.

Pneumothorax

- **Simple:** hyperresonance, ↓breath sounds. Small/occult → observe; moderate/large → **chest tube (5th ICS, anterior axillary, safe triangle).**
- **Open ("sucking chest wound"):** chest wall defect. → **three-way occlusive dressing** (prevents tension) → chest tube → definitive closure.
- **Tension (EMERGENCY):** one-way valve → mediastinal shift. **Distended neck veins, hypotension, tracheal deviation to contralateral side, absent breath sounds, hyperresonance.**

- **CLINICAL diagnosis (before CXR).** Immediate needle decompression: 2nd intercostal space, mid-clavicular line, 16G → then chest tube.

Haemothorax

- Blood in pleura. **Lung = commonest source (self-limiting)**; intercostal/internal mammary/hilar = ongoing.
- ↓breath sounds + **dullness to percussion**. X-ray >200mL.
- **Chest tube.** Thoracotomy if: initial >1500mL OR >200mL/h for 2–3h OR persistent transfusion need.

Chest tube (ICD)

- **5th intercostal space, anterior axillary line — "safe triangle".** Blunt dissection over top of rib. Underwater seal = one-way valve; bottle below thorax.
- Indications: pneumothorax (tension/spontaneous/iatrogenic), haemothorax, haemopneumothorax, empyema, chylothorax, bronchopleural fistula.

Tracheobronchial injury

- MVC; **80% within 2.5cm of carina**; massive air leak, haemoptysis, sub-cut emphysema.
- **Persistent large pneumothorax + continuous air leak after chest tube = clue.**
- **Bronchoscopy = definitive diagnosis.** Intubation beyond injury; surgical repair. Mortality 10%.

Cardiovascular injuries

- **Myocardial contusion:** >55% of severe closed chest trauma. Monitor; **ECG + serial troponin** (normal at 4–6h excludes).
- **Penetrating cardiac: RV most common (43%; LV 34%);** suspect in **"the BOX"** (clavicles–mid-clavicular lines–costal margins). 80% die at scene.
- **Cardiac tamponade: Beck's triad = muffled heart sounds + distended neck veins + hypotension;** narrow pulse pressure, pulsus paradoxus.
- **ECHO** (effusion + RV collapse); ECG electrical alternans. Fluids; **pericardiocentesis** (blunt/stable); **thoracotomy + repair** (penetrating).
- **Blunt aortic injury: most common vessel injured by blunt trauma. 80–90% at descending aorta distal to left subclavian (isthmus).**
- CXR: **widened mediastinum >8cm**, loss of aortic knob, apical cap, L effusion.
- **CT angiography = diagnosis.** Fix ABCDE first; **control BP (SBP 100–120, β-blocker/esmolol)**; surgery or endovascular repair. 85% survive if prompt.

Oesophageal perforation

- Penetrating; high morbidity, mediastinitis. **Gastric content in chest tube**, pneumomediastinum, effusion without chest injury.

Diaphragmatic tear

- Severe trauma + associated injuries → **mandates abdominal exploration.**

ED thoracotomy

- **Indications:** cardiac arrest in **PENETRATING** injury, massive haemothorax, penetrating + tamponade, large open wound, major vascular/tracheobronchial injury, oesophageal perforation. **CONTRAINDICATED in pulseless BLUNT trauma.**

- **5 moves:** 1. incise inferior pulmonary ligament · 2. open pericardium + staple/suture cardiac laceration · 3. open cardiac massage · 4. clamp pulmonary hilum / twist bleeding lung · 5. **clamp thoracic aorta.**

Thoracic trauma — high-yield

- **Tension pneumo** = clinical dx → needle decompress (2nd ICS, mid-clavicular) → chest tube (5th ICS, safe triangle).
- **Open pneumo** = 3-way occlusive dressing → tube.
- **Haemothorax** thoracotomy: >1500mL or >200mL/h.
- **Flail** = ≥ 3 ribs $\times 2$ places, paradoxical, treat contusion.
- **Lower rib #** → liver/spleen/kidney. 1st/2nd rib + cap → great vessel.
- **Blunt aorta:** distal to L subclavian; widened mediastinum; CT angio; β -blocker.
- **Tamponade** = Beck's triad; ECHO; pericardiocentesis. Penetrating = "the BOX", RV most common.
- **Tracheobronchial:** persistent air leak after tube → bronchoscopy.
- **ED thoracotomy:** penetrating arrest yes, blunt arrest no.

PERITONEAL CAVITY & RETROPERITONEUM (IMAGING)

Core principle

- **CT = mainstay. US + MRI = no ionizing radiation.** Interpretation = anatomy + pathology principles.

Peritoneal cavity — ascites

- Small amounts detected US/CT/MRI; **nature NOT reliably differentiated.**
- US detects smallest amount. CT = low density.
- **Free ascites** → **dependent sites: pouch of Douglas, Morrison's pouch (hepatorenal), paracolic gutters.** Loculated → mimics abscess.

Peritoneal tumours

- Mostly metastases — **ovarian origin most important.**
- Imaging: (a) ascites only, (b) **peritoneal nodules**, (c) **omental cake.**

Intra-peritoneal abscess

- Follows **perforation**; site depends on perforation site; multiple common; ascites frequent.
- **US:** loculated fluid + irregular wall, internal echoes (septation/debris), air; DDx bowel (peristalsis).
- **CT:** fluid density + enhancing wall, **air in 50%** (bubble / streak / air-fluid level).
- **In-111/Tc-labelled leukocyte scan** if US/CT fail (e.g. anastomotic leak).

Retroperitoneum — anatomy

- **3 compartments** (anterior + posterior renal fascia). Organs: adrenal, aorta, retrocrural, IVC, psoas. **CT main**; plain film (mass/calcification/gas); MRI (multiplanar, vascular extension).

Retroperitoneal mass — DDx (7)

- 1. **Nodal** (lymphoma, germ cell, metastases)
- 2. Retroperitoneal fibrosis
- 3. **Neurogenic** (paraganglioma, schwannoma, ganglioneuroma)
- 4. **Sarcoma** (liposarcoma, fibrosarcoma)
- 5. **Haematoma** (trauma, ruptured AAA)
- 6. Infection (psoas/retroperitoneal abscess)
- 7. Vascular (aneurysm)

Lymphadenopathy

- Metastases & lymphoma look **identical** → **FDG-PET/CT** (esp lymphoma follow-up).
- Normal short axis: **para-aortic 10mm, retrocrural 6mm. >20mm = invariably malignant.**
- Lymphoma → enlarged nodes → confluent lobulated mass around aorta.

Adrenal imaging (CT best routine)

- **Non-functioning adenoma:** <3cm hard to tell from mets; >3cm = mostly mets (rarely carcinoma). Metastases common, **frequently bilateral, lung common primary.**
- **Cushing's adenoma:** always >2cm, always seen, <10 HU.
- **Aldosteronoma:** usually <1cm — difficult to identify.
- **Hyperplasia:** normal/mild enlargement.

- **Phaeochromocytoma:** very large, all detected; **10% rule (multiple, bilateral, extra-adrenal, malignant); MIBG scan** localizes.

Retroperitoneal tumours

- Mainly liposarcoma/fibrosarcoma; nature usually undeterminable.
- **Liposarcoma recognizable = fat + strands of soft tissue density.**

Aortic aneurysm (AAA)

- **US = screening;** normal 2.5cm, **>3cm = aneurysm.** CT + US = true dimensions. Plain film lateral = calcification.
- CT: blood clot lining wall, calcification, IVC displacement.
- **>6cm = high rupture risk. Leaking aneurysm easily diagnosed on CT** (haemorrhage in retroperitoneum).

Imaging — high-yield

- **CT mainstay; US/MRI no radiation.**
- **Ascites:** pouch of Douglas, Morrison's pouch, paracolic gutters. Peritoneal mets → omental cake (ovarian).
- **Abscess:** air in 50% on CT; leukocyte scan if CT/US fail.
- **Node >20mm = malignant** (para-aortic 10mm); lymphoma ≈ mets on size → **FDG-PET.**
- **Cushing's adenoma >2cm (<10HU); aldosteronoma <1cm; phaeo 10% rule + MIBG.**
- **Liposarcoma = fat + soft tissue strands.**
- **AAA: US screen, >3cm aneurysm, >6cm rupture risk, CT for leak.**

THORACIC TRAUMA — Clinical Revision Table

Bold = high-yield. 2nd leading cause trauma death; 25% of trauma deaths from chest injury. Early deaths often PREVENTABLE (tension pneumo, tamponade, airway, haemorrhage).

CHEST WALL			
Injury	Key features / signs	Management	Pearls
Rib fractures	Heal 3–6 wks. Ribs 4–9 commonest. Lower ribs (X–XII) → liver/spleen/kidney. 1st/2nd # + apical cap → great vessel injury	Pain control, pulmonary function. Repeat CXR 3h (delayed haemo/pneumo). Elderly + multiple # → admit (pneumonia)	Splinting → atelectasis/pneumonia
Flail chest	≥3 ribs in ≥2 places → free segment → PARADOXICAL respiration (in on inspiration). Associated pulmonary contusion (real problem)	Pain control, pulmonary toilet, O2, treat contusion. Severe → intubation + IPPV (internal pneumatic fixation)	Contusion + hypoxaemia kill, not the flail
Sternal fracture	Anterior blunt / seatbelt deceleration . Low mortality. → myocardial contusion risk	Lateral X-ray; screen myocardial contusion (ECG + serial cardiac enzymes), monitor 24–48h ; analgesia	NO association with blunt aortic injury
Subcutaneous emphysema	Air in chest wall → marker of serious injury	Benign + self-limiting → high-flow O2 . Massive → 'gills' incisions	—

PLEURAL / LUNG			
Injury	Key features / signs	Management	Pearls
Pulmonary contusion	Alveolar oedema/haemorrhage; hypoxaemia ~50%; X-ray findings within minutes, ARDS-like	Restrict IV fluids , pulmonary toilet, pain control, respiratory support. NO prophylactic antibiotics . Mortality 5–16%	Worsens over 24–48h
Simple pneumothorax	Air in pleural space; hyperresonance, ↓breath sounds. Grades: small/large/occult	Small/occult → observe. Moderate/large → chest tube (5th ICS, anterior axillary, safe triangle)	—
Open pneumothorax	'Sucking chest wound' ; chest wall defect; respiratory distress	Three-way occlusive dressing (prevents tension) → chest tube → definitive closure	Penetrating > blunt
Tension pneumothorax	EMERGENCY. One-way valve → mediastinal shift. Distended neck veins, hypotension, tracheal deviation (contralateral), absent breath sounds, hyperresonance	CLINICAL dx (before CXR). Immediate needle decompression: 2nd ICS, mid-clavicular line, 16G → then chest tube	↓venous return → shock
Haemothorax	Blood in pleura (lung = commonest source, self-limiting). ↓breath sounds + DULLNESS . X-ray >200mL	Chest tube. Thoracotomy if: initial >1500mL OR >200mL/h x2–3h OR persistent transfusion	Massive = up to 3L

AIRWAY / OESOPHAGUS / DIAPHRAGM			
Injury	Key features / signs	Management	Pearls
Tracheobronchial injury	MVC; 80% within 2.5cm of carina ; massive air leak, haemoptysis, sub-cut emphysema. Persistent large pneumo + air leak after chest tube	Bronchoscopy = definitive dx . Intubation beyond injury; surgical repair (thoracotomy)	Mortality 10%
Oesophageal perforation	Penetrating; high morbidity, mediastinitis. Gastric content in chest tube , pneumomediastinum, effusion without chest injury	Surgical repair; antibiotics	2nd to penetrating injury
Diaphragmatic tear	Severe trauma + associated injuries	Mandates abdominal exploration	—

CARDIOVASCULAR			
Injury	Key features / signs	Management	Pearls
Myocardial contusion	>55% of severe closed chest trauma	Monitor; ECG + serial troponin (normal at 4–6h excludes); O2, analgesia	Risk stratify with troponin
Penetrating cardiac / tamponade	RV most common (43%; LV 34%) ; suspect in 'the BOX' (clavicles–mid-clavicular–costal margins). Beck's triad: muffled heart sounds + distended neck veins + hypotension ; narrow pulse pressure, pulsus paradoxus	ECHO (effusion + RV collapse); ECG electrical alternans. Fluids; pericardiocentesis (blunt/stable); thoracotomy + repair (penetrating)	80% die at scene
Blunt aortic injury	Most common vessel injured by blunt trauma. 80–90% at descending aorta distal to left subclavian (isthmus) . CXR: widened mediastinum >8cm , loss of aortic knob, apical cap, L effusion	CT angiography (helical) = dx . Fix ABCDE first; control BP (SBP 100–120, β-blocker/esmolol) ; surgery or endovascular repair	85% survive if prompt; paraplegia 5–7%

EMERGENCY DEPARTMENT THORACOTOMY	
Indications	Cardiac arrest in PENETRATING injury, massive haemothorax, penetrating + tamponade, large open wound, major vascular/tracheobronchial injury, oesophageal perforation. CONTRAINDICATED in pulseless BLUNT trauma

5 moves

1. Incise inferior pulmonary ligament · 2. Open pericardium + staple/suture cardiac laceration · 3. Open cardiac massage · 4. Clamp pulmonary hilum / twist bleeding lung · 5. **Clamp thoracic aorta**

PERITONEAL CAVITY & RETROPERITONEUM — Imaging Revision Table

Bold = high-yield. CT = mainstay; US/MRI = no ionizing radiation. Radiology lecture (Dr Kawa Mahmood).

PERITONEAL CAVITY

Condition	Cause	Imaging findings
Ascites	Small amounts detected US/CT/MRI; nature NOT reliably differentiated	US detects smallest amount. CT = low density. Free → dependent sites: pouch of Douglas, Morrison's pouch, paracolic gutters. Loculated → mimics abscess
Peritoneal tumours	Mostly metastases — ovarian most important	(a) ascites only, (b) peritoneal nodules , (c) omental cake
Intra-peritoneal abscess	Follows perforation ; site depends on perforation; multiple common; ascites frequent	US : loculated + irregular wall, internal echoes, air; DDx bowel (peristalsis). CT : fluid + enhancing wall, air in 50% (bubble/streak/air-fluid level). In-111/Tc leukocyte scan if US/CT fail (e.g. anastomotic leak)

RETROPERITONEUM — ANATOMY & MASSES

Anatomy / modalities	3 compartments (anterior + posterior renal fascia). Organs: adrenal, aorta, retrocrural, IVC, psoas. CT main ; plain film (mass/calcification/gas); MRI (multiplanar, vascular extension)
Mass DDx (7)	1. Nodal (lymphoma, germ cell, mets) 2. Retroperitoneal fibrosis 3. Neurogenic (paraganglioma, schwannoma, ganglioneuroma) 4. Sarcoma (lipo/fibro) 5. Haematoma (trauma/ruptured AAA) 6. Infection (psoas abscess) 7. Vascular (aneurysm)
Lymphadenopathy	Mets & lymphoma look identical → FDG-PET/CT (esp lymphoma follow-up). Normal short axis: para-aortic 10mm, retrocrural 6mm; >20mm = invariably malignant. Lymphoma → confluent lobulated mass around aorta
Retroperitoneal tumours	Mainly liposarcoma/fibrosarcoma; nature usually undeterminable. Liposarcoma recognizable = fat + strands of soft tissue density

ADRENAL IMAGING (CT best routine)

Non-functioning adenoma	<3cm hard to tell from mets; >3cm = mostly mets (rarely carcinoma). Metastases common, frequently bilateral, lung common primary
Cushing's adenoma	Always >2cm, always seen CT/MRI; <10 HU density
Aldosteronoma	Usually <1cm — difficult to identify
Hyperplasia	Normal / mild enlargement
Phaeochromocytoma	Very large, all detected; 10% rule (multiple, bilateral, extra-adrenal, malignant); MIBG scan for localization

AORTIC ANEURYSM (AAA)

Screening / dimensions	US = screening ; normal 2.5cm, >3cm = aneurysm. CT + US = true dimensions. Plain film lateral = calcification. Angiography ≠ true size (good for branch stenosis)
Findings / rupture	CT: blood clot lining wall, calcification, IVC displacement. >6cm = high rupture risk. Leaking/bleeding aneurysm easily diagnosed on CT (haemorrhage in retroperitoneum)